

CHAPTER 2 : HEALTH AND DISEASES

Topic:-

Prepare posters giving information about various diseases, public awareness and arrange exhibition in school.

Introduction:-

Health is defined by the World Health Organization (WHO) as a complete state of physical, mental, and social well-being, and not merely the absence of disease.

Diseases cause discomfort, pain, and prevent our organs from functioning smoothly. In science, diseases are broadly divided into infectious (communicable) diseases and non-infectious diseases.

Infectious diseases like Dengue, Malaria, Typhoid, and Tuberculosis spread from person to person through contaminated air, water, food, or insect bites. Non-infectious diseases like Diabetes, High Blood Pressure, and Cancer do not spread from person to person, but develop due to unhealthy lifestyle habits, lack of physical activity, or genetic factors. Creating educational posters and organizing an exhibition in school is an active way for students to spread health awareness.

❖ Two Main Categories of Diseases:-

- **Infectious Diseases:** Spread rapidly by microbes and vectors (Dengue, TB, Flu).
- **Non-Infectious Diseases:** Arise from lifestyle and diet (Diabetes, Heart troubles).
- **Key Goal:** Educate fellow students and parents on prevention and early care.

Aim & Exhibition Preparation:-

Aim: To design attractive, informative health posters on infectious and lifestyle diseases, and organize a public awareness exhibition in our school hall.

❖ Objectives of the Exhibition:-

- To illustrate causes, symptoms, and prevention of major diseases using diagrams.
- To demonstrate the breeding cycle of mosquitoes and how to eliminate them.
- To explain healthy eating habits, regular exercise, and yoga to school children.

❖ Materials Required:-

- Colored drawing sheets, acrylic colors, marker pens, sketch pens, and scissors.
- Display display boards, pins, double-sided tape, and display stands.
- Class 8 General Science textbook (Chapter 2: Health and Diseases).



Procedure & Organizing the Exhibition:-

We organized the school health exhibition by following these organized steps:

- **Step 1: Forming Student Teams:** Our class was divided into 4 teams covering: Vector-Borne Diseases, Water-Borne Diseases, Lifestyle Diseases, and Mental Health.
- **Step 2: Poster Designing:** Each team created bold posters with clear slogans, colorful illustrations, and easy bullet points in simple language.
- **Step 3: Setting Up Exhibition Stalls:** We decorated the school auditorium with themed display boards, banners, and sample mosquito larval water bowls.
- **Step 4: Demonstrations & Speeches:** Student volunteers explained poster concepts to visiting students, teachers, and parents.
- **Step 5: Handing Out Health Leaflets:** We printed simple DIY health checklists to give to visiting parents on home mosquito prevention.

❖ Key Exhibition Slogans Created:-

- *"Clean Water, Green Surrounding, Mosquito Free Living!"*
- *"An Apple a Day and 30 Minutes of Play Keeps the Doctor Away!"*

Observations & Poster Details Table:-

The table below lists the posters prepared and the core message displayed in the exhibition:

<i>Poster Theme</i>	<i>Disease Highlighted</i>	<i>Main Causes Shown</i>	<i>Key Preventive Message</i>
Vector Menace	Dengue & Malaria	Aedes / Anopheles mosquitoes in stagnant water	Empty coolers weekly; observe Sunday Dry Day
Airborne Germs	Tuberculosis (TB)	Mycobacterium bacilli via coughing droplets	Wear masks, do not spit openly, take BCG
Lifestyle Illness	Diabetes (Type 2)	Excess sugar, obesity, lack of exercise	Walk 5,000 steps daily, eat green vegetables
Cardiac Health	Heart Attacks & BP	Deep fried food, mental stress, smoking	Practice pranayama, reduce salt and oil intake
Nutrition Defense	Anemia & Malnutrition	Lack of iron, vitamins, skipping meals	Eat jaggery, spinach, lentils, and fresh fruits

Scientific Analysis & Discussion:-

During the exhibition, we discussed several deep scientific concepts with visitors:

- **Why Mosquito Vectors Are Dangerous:** Mosquitoes act as vectors, meaning they carry pathogens from a sick patient to a healthy person without falling sick themselves. Aedes mosquitoes bite during daytime and lay eggs in clean domestic water (coolers, flower pots, discarded tyres).
- **The Modern Epidemic of Lifestyle Diseases:** Unlike our grandparents who did plenty of physical farming work, today many children spend long hours watching mobile screens and eating packaged chips. This leads to early childhood obesity and insulin resistance (Diabetes).
- **Community Immunity & Herd Protection:** When all students in a school are immunized and observe personal cleanliness, disease outbreaks cannot spread across the community.

❖ Feedback from Visiting Parents:-

Over 250 parents visited our stalls. Many parents admitted they did not know that clean cooler water could breed dangerous dengue mosquitoes!

Everyday Applications & Impact:-

The school health exhibition created real positive changes in our homes and school:

- **Weekly Dry Day at Home:** Students promised to empty, scrub, and dry all water coolers and pots every Sunday to break mosquito lifecycles.
- **Healthy School Tiffin Drive:** Our school banned oily packaged junk chips in the cafeteria and promoted fresh fruit tiffins.
- **Clean Classroom Drive:** Dustbins with lids were placed in all classrooms, and student hygiene monitors were appointed.

❖ Major Advantages of Health Exhibitions:-

- Transforms abstract textbook science into practical lifesaving daily habits.
- Builds public speaking skills and scientific confidence in school students.
- Directly educates the wider neighborhood through child-to-parent learning.

❖ Limitations Identified:-

Posters alone cannot eliminate diseases unless local municipal corporations maintain clean covered drainage and provide reliable piped drinking water.

What We Learned & Conclusion:-

- Health is a total state of physical and mental well-being; prevention is always better than cure.
- Infectious diseases can be stopped by eliminating insect vectors and practicing hygiene.
- Non-infectious lifestyle diseases are managed through active exercise, healthy food, and low stress.
- Visual charts and exhibitions are powerful educational tools to build healthy communities.

Conclusion:-

Our school health poster exhibition was a tremendous success. It proved that young students can become effective health champions in society. By sharing scientific knowledge about diseases, vectors, and nutrition through creative art, we empowered our school community to lead healthier, happier, and disease-free lives.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 2: Health and Diseases).
- World Health Organization (WHO) – Health Promotion in Schools Guidelines.
- Public Health Department, Government of Maharashtra Awareness Manual.